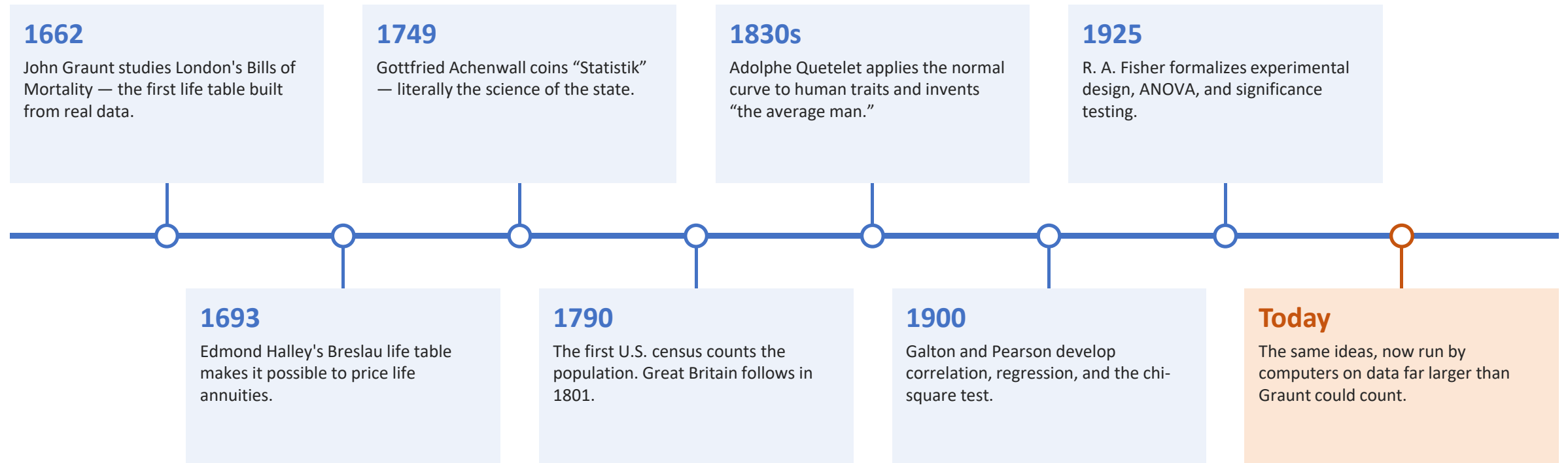


Lecture 1

Introduction to Statistical Methods

Where does statistics come from?

Statistics began as bookkeeping about people — counting births, deaths, and taxpayers so that a state could describe itself.



Sources: J. Graunt (1662), *Natural and Political Observations Made upon the Bills of Mortality*. E. Halley (1693), *Phil. Trans. R. Soc.* 17:596–610. G. Achenwall (1749), *Abriss der neuesten Staatswissenschaft*. A. Quetelet (1835), *Sur l'homme et le développement de ses facultés*. R. A. Fisher (1925), *Statistical Methods for Research Workers*.

Answering Questions with data

- What proportion of people in the U.S are biologically female?
Survey Americans and ask them to report their sex
- Does a low-carb diet result in significant weight loss?
 - Design an experiment to evaluate the effectiveness of the low carb diet on weight loss
 - Record information such as starting and ending weight, calories consumed per day, ...
- Are people more likely to stop at Starbucks if they've recently seen a Starbucks TV add?
 - Conduct a marketing survey to record the number of people who have gone to starbucks since add aired
 - Compare between those who saw the add and those who did not



Where do we find Statistics? – Everywhere!

- **Sports** (number of points scored per game by an athlete)
- **Weather** Forecasting (average monthly rainfall)
- **Economy** (median income, unemployment rate)
- **Sales Tracking** (projected sales revenue from iphone 14's sold in a given month)
- **Medicine** (percentage of people who were pain free from using a drug)
- **Manufacturing** (number of cans of coke produced in a given month)

The Science of Data

Statistical science deals with the collection, analysis, interpretation, and presentation of data

Data is a collection of observations/measurements, and can be **structured** or **unstructured**.

Structured data

Predefined format: fixed rows, columns and fields, easily stored in databases.

Examples:

- Spreadsheets and CSV files
- Survey responses, sensor readings
- Transaction records

Unstructured data

No predefined format or fixed fields; requires processing before it can be analyzed.

Examples:

- Free-text notes, emails, social posts
- Images, audio and video
- Web pages and PDF documents

What Do Data Look Like?

The following table records sales of child car seats from 20 stores. For each store, the unit sales, price charged, community income, regional population, local education level, shelving quality, and whether the store is urban and in the U.S. are recorded

Reading a data table

Each column is a variable

Sales	Price	Income	ShelveLoc	Urban
4.38	108	98	Bad	Yes
11.62	139	83	Good	Yes
5.07	96	91	Bad	Yes
6.87	131	105	Medium	Yes

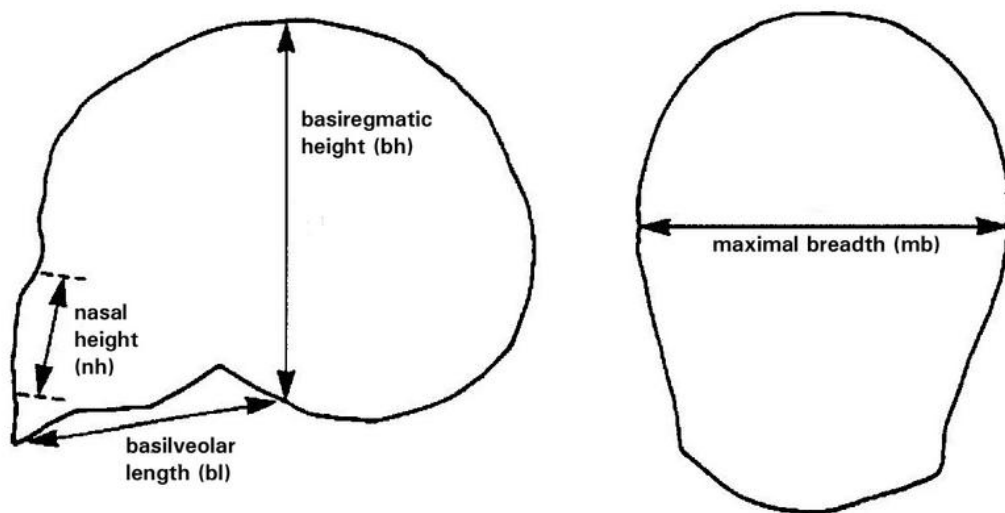
Each row is one observation — one store

Observation	Sales (1,000s)	Price (\$)	Income (\$1,000s)	Population (1,000s)	Education (years)	ShelveLoc	Urban	US
1	4.38	108	98	173	16	Bad	Yes	No
2	11.62	139	83	325	17	Good	Yes	Yes
3	5.07	96	91	334	17	Bad	Yes	Yes
4	6.87	131	105	249	13	Medium	Yes	Yes
5	11.27	133	68	60	16	Good	Yes	Yes
6	3.67	131	31	327	16	Medium	Yes	No
7	4.19	112	93	420	11	Bad	Yes	Yes
8	4.97	160	67	27	17	Medium	Yes	Yes
9	4.36	123	41	357	14	Bad	No	Yes
10	8.69	101	64	68	16	Medium	Yes	Yes
11	8.65	120	76	218	14	Medium	No	Yes
12	9.16	156	105	435	14	Good	Yes	Yes
13	6.37	132	77	86	18	Medium	Yes	Yes
14	4.21	137	35	502	10	Medium	No	Yes
15	8.57	99	78	158	11	Medium	Yes	Yes
16	6.80	135	117	337	10	Bad	Yes	Yes
17	8.61	102	38	283	15	Medium	Yes	No
18	10.26	108	75	377	12	Good	Yes	No
19	7.90	124	46	206	11	Medium	Yes	No
20	6.81	125	61	263	12	Medium	No	No

Source: Carseats data set, R package ISLR2. James, G., Witten, D., Hastie, T., and Tibshirani, R. (2021). An Introduction to Statistical Learning, 2nd ed. Springer. Carseats is a simulated data set of 400 stores distributed with the textbook; 20 stores are shown here.

Example data 2

The following are data from an analysis of 150 Egyptian skulls from 5 epochs of Egyptian history. For each skull, the epoch, and several measurements characterizing the shape of the skull are recorded



Source: D. J. Hand, F. Daly, A. D. Lunn, K. J. McConway and E. Ostrowski (1994). A Handbook of Small Datasets, Chapman and Hall/CRC, London.

Visualizing Tests for Equality of Covariance Matrices - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Diagram-of-the-skull-measurements-for-the-Egyptian-skulls-data-set-Maximal-breadth-and_fig4_325126938 [accessed 21 Aug, 2023]

Observation	epoch	mb	bh	bl	nh
1	c4000BC	131	138	89	49
2	c4000BC	125	131	92	48
3	c4000BC	131	132	99	50
4	c4000BC	119	132	96	44
5	c4000BC	136	143	100	54
6	c4000BC	138	137	89	56
7	c4000BC	139	130	108	48
8	c4000BC	125	136	93	48
9	c4000BC	131	134	102	51
10	c4000BC	134	134	99	51
11	c4000BC	129	138	95	50
12	c4000BC	134	121	95	53
13	c4000BC	126	129	109	51
14	c4000BC	132	136	100	50
15	c4000BC	141	140	100	51
⋮	⋮	⋮	⋮	⋮	⋮
145	cAD150	132	127	97	52
146	cAD150	137	125	85	57
147	cAD150	129	128	81	52
148	cAD150	140	135	103	48
149	cAD150	147	129	87	48
150	cAD150	136	133	97	51

Checkpoint:



Data - a collection of observations/measurements on a set of variables - typically represented as a table.

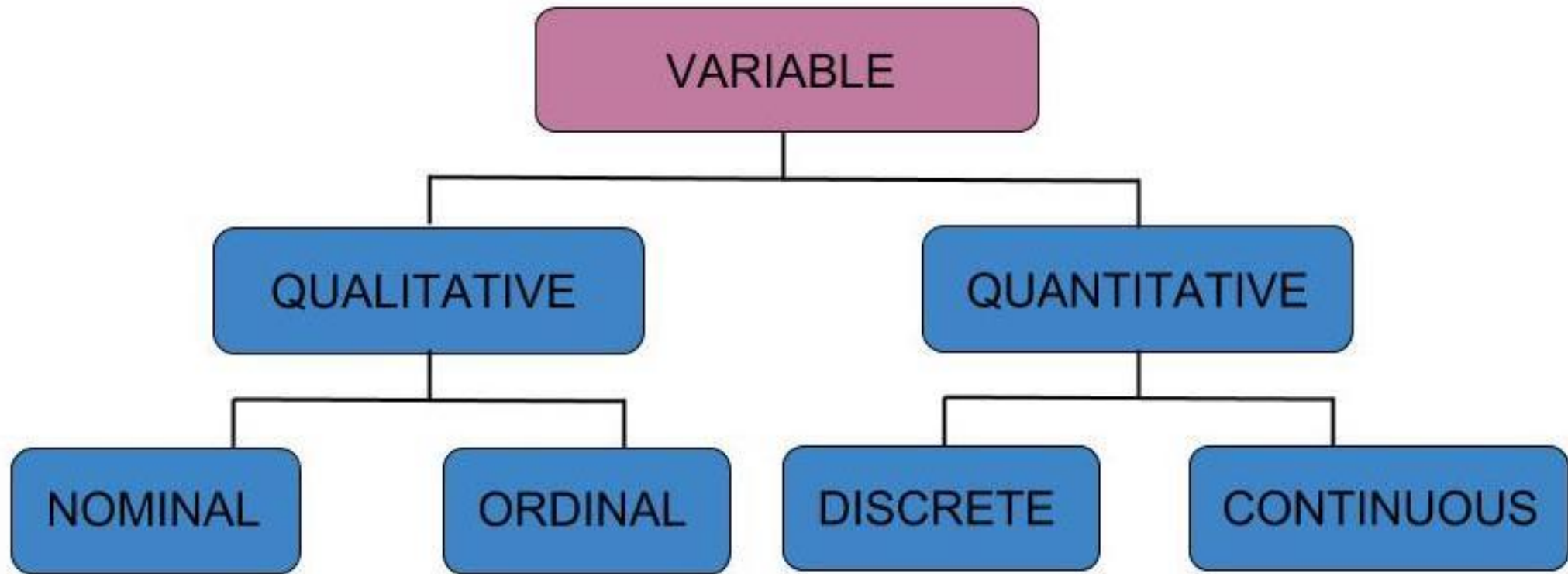


Observation – the fundamental unit of data – typically observations are rows of a data table



Variables – characteristics of an observation – typically variables are the columns of a data table

Types of Variables: Qualitative vs Quantitative



Types of Variables: Qualitative vs Quantitative

Qualitative (Categorical) variable – non-numeric qualities or characteristics that can be placed in distinct categories

State/city (ID, WA, MT, ...)

Treatment (Drug/Placebo)

Genotype (AA, AT, TT)

Survival (live or die)



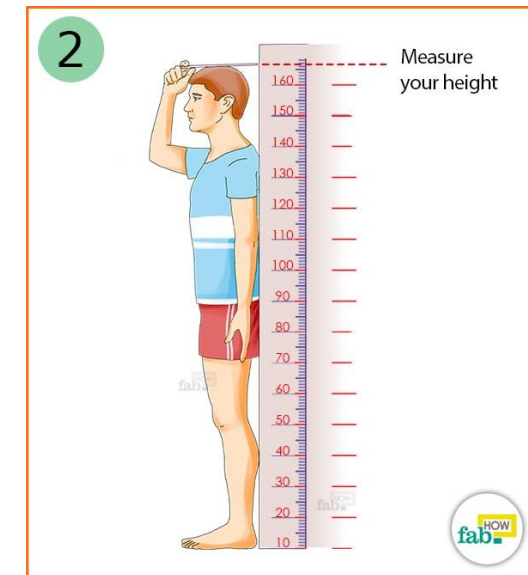
Quantitative (numeric) variable – numerical characteristics - can be ordered or ranked

Height (inches/cm)

Weight (lbs/kg)

Longevity/Age (number of years)

Dose (micrograms per gram)



Practice: Qualitative vs Quantitative

From our example data of egyptian skulls, which variables are qualitative and which are quantitative?

Qualitative: Epoch

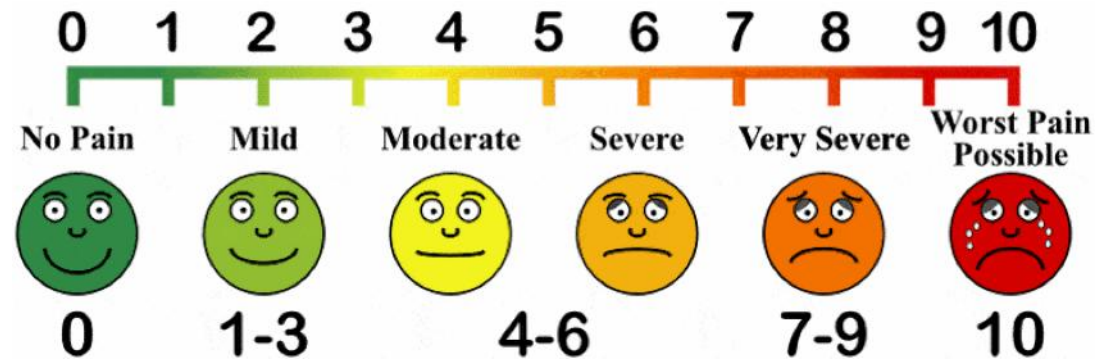
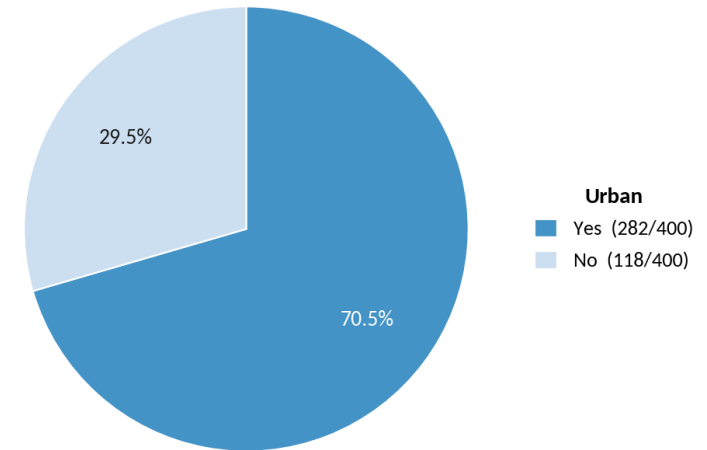
Quantitative: mb, bh, bl, nh

Observation	epoch	mb	bh	bl	nh
1	c4000BC	131	138	89	49
2	c4000BC	125	131	92	48
3	c4000BC	131	132	99	50
4	c4000BC	119	132	96	44
5	c4000BC	136	143	100	54
6	c4000BC	138	137	89	56
7	c4000BC	139	130	108	48
8	c4000BC	125	136	93	48
9	c4000BC	131	134	102	51
10	c4000BC	134	134	99	51
11	c4000BC	129	138	95	50
12	c4000BC	134	121	95	53
13	c4000BC	126	129	109	51
14	c4000BC	132	136	100	50
15	c4000BC	141	140	100	51
⋮	⋮	⋮	⋮	⋮	⋮
145	cAD150	132	127	97	52
146	cAD150	137	125	85	57
147	cAD150	129	128	81	52
148	cAD150	140	135	103	48
149	cAD150	147	129	87	48
150	cAD150	136	133	97	51

Types of Variables: Qualitative vs Quantitative

Qualitative nominal – non-numeric qualities or characteristics that can be placed in distinct categories that do not have a natural ordering (e.g labels, names, colors, etc)

Qualitative ordinal – non-numeric qualities or characteristics that can be placed in distinct categories with an inherent ordering (Likert responses, education level, military ranking)



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Practice: Qualitative vs Quantitative

In the Carseats example data, which variable(s) are qualitative nominal and which are qualitative ordinal?

Qualitative nominal: Urban, US

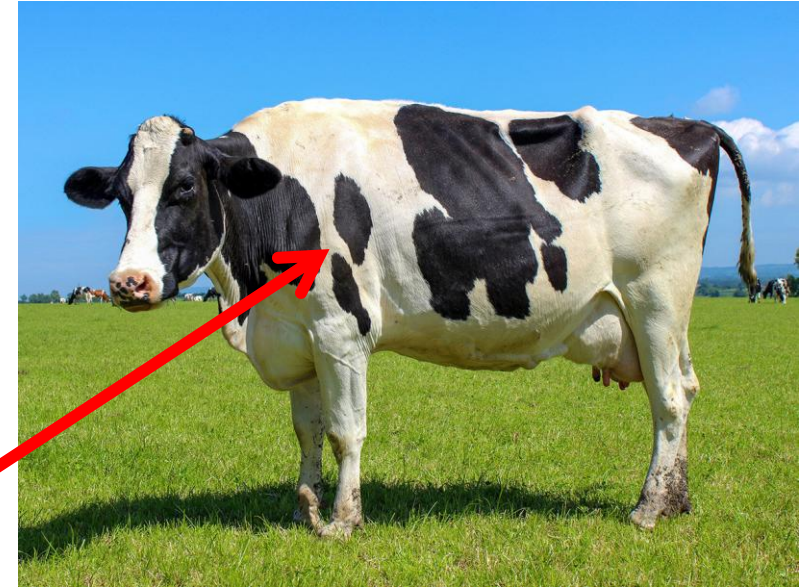
Qualitative ordinal: ShelfLoc
(Bad < Medium < Good)

Observation	Sales (1,000s)	Price (\$)	Income (\$1,000s)	Population (1,000s)	Education (years)	ShelveLoc	Urban	US
1	4.38	108	98	173	16	Bad	Yes	No
2	11.62	139	83	325	17	Good	Yes	Yes
3	5.07	96	91	334	17	Bad	Yes	Yes
4	6.87	131	105	249	13	Medium	Yes	Yes
5	11.27	133	68	60	16	Good	Yes	Yes
6	3.67	131	31	327	16	Medium	Yes	No
7	4.19	112	93	420	11	Bad	Yes	Yes
8	4.97	160	67	27	17	Medium	Yes	Yes
9	4.36	123	41	357	14	Bad	No	Yes
10	8.69	101	64	68	16	Medium	Yes	Yes
11	8.65	120	76	218	14	Medium	No	Yes
12	9.16	156	105	435	14	Good	Yes	Yes
13	6.37	132	77	86	18	Medium	Yes	Yes
14	4.21	137	35	502	10	Medium	No	Yes
15	8.57	99	78	158	11	Medium	Yes	Yes
16	6.80	135	117	337	10	Bad	Yes	Yes
17	8.61	102	38	283	15	Medium	Yes	No
18	10.26	108	75	377	12	Good	Yes	No
19	7.90	124	46	206	11	Medium	Yes	No
20	6.81	125	61	263	12	Medium	No	No

Quantitative Variables: Discrete vs Continuous

- **Quantitative discrete** – quantitative variables that take on distinct, countable values such 0,1,2,3... (whole numbers or integers)
 - all counts are quantitative discrete variables

e.g. number of black spots on a dairy cow



- **Quantitative continuous** – variables that can take on infinite number of values within an interval of any two specific values (e.g. temperature °F, height in inches, speed in miles per hour)



Practice: Discrete vs Continuous

In the Carseats example data, which variable(s) are quantitative discrete and which are quantitative continuous?

Quantitative discrete:
Population, Education (both counts)

Quantitative continuous: Sales, Price, Income

Observation	Sales (1,000s)	Price (\$)	Income (\$1,000s)	Population (1,000s)	Education (years)	ShelveLoc	Urban	US
1	4.38	108	98	173	16	Bad	Yes	No
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